0107571 - Internet Programming - Take Home Final Exam

Department of Computer and Communications Engineering
Tafila Technical University

Due on Monday 31-8-2026 at 11:30, Summer 2025-2026

Problem 1. Design and implement a full-stack web application for a **domain/topic of your choice** (e.g., E-Commerce Store, Online Library, Course Booking, Healthcare Portal, Event Management, etc.) that meets the following requirements: **(36 pts)**

1. **User Roles & Navigation:** On the main page, a user must be able to navigate to at least four distinct sections representing key system roles/views (e.g., Customer/Client, Provider/Service, About/Information, Administrator).
2. **About Page:** Display detailed background information and features of the selected website domain.
3. **Client/Customer Interface:** Allow end-users to view items/services and perform specific selections or reservations (e.g., selecting specific products, booking services, or reserving specific seats/slots).
4. **Provider/Driver Interface:** Provide a dedicated dashboard where service providers or drivers can view assigned customers, orders, or reservation details alongside their relevant location/contact data.
5. **Admin Dashboard (CRUD):** On the administrator page, implement full CRUD functionality allowing the admin to **C**reate tables/records, **R**ead existing data, **U**pdate records, and **D**elete rows or entities.
6. **Database Architecture:** Design and implement a MySQL database structure with properly normalized relational tables appropriate for your chosen topic (e.g., storing entities such as users, transactions, items, and bookings).
7. **Search Functionality:** Incorporate a functional search bar within the main client and administrator pages to filter and search records (e.g., trips, products, or orders).
8. **Dynamic Pagination:** Implement pagination to traverse database records whenever list results exceed 5 items, utilizing dynamic AJAX/asynchronous loading techniques.
9. **Authentication System:** Include functional Login, Logout, and Registration mechanisms accessible across the site for users, providers, and administrators.
10. **User State Visibility:** Display a dynamic login indicator (e.g., user icon, active username, or profile badge) at the top of every page when an entity is authenticated.
11. **Consistent Layout:** Ensure a unified layout featuring a persistent site header and footer across all views.
12. **Responsive Design:** Utilize modern CSS techniques (Flexbox, Grid, or Media Queries) to ensure full mobile and desktop responsiveness.

13. **Dynamic Page Loading:** Apply asynchronous loading techniques (e.g., AJAX/Fetch API) when fetching page content or switching sections without full page refreshes.
14. **Branding & Assets:** Maintain a custom logo in the site header across all pages.
15. **Favicon & Metadata:** Configure an appropriate favicon alongside dynamic HTML page titles for each view.
16. **Session Management:** Utilize PHP session arrays ('$_SESSION') to persist user state across different browser tabs and requests.
17. **PRG Pattern:** Implement the Post/Redirect/Get (PRG) pattern on all form submissions to prevent accidental data resubmission upon page refresh.
18. **Web Security:** Enforce strict security measures in PHP and JavaScript to prevent Cross-Site Scripting (HTML injection via 'htmlspecialchars()') and SQL Injections (using Prepared Statements).

Problem 2. Be prepared to demonstrate your live application and answer technical questions regarding your backend architecture, database schema, and frontend code during the oral defense on **the day of the final exam**. **(4 pts)**